

# Dan Sapp

Senior Software Engineer

St. Paul, MN | [business@dan-sapp.com](mailto:business@dan-sapp.com) (preferred) | (763)-218-2340

[LinkedIn](#) | [GitHub](#) | [Real-time 3D AIS and ADS-B Telemetry Platform](#)

Software engineer with 10+ years of experience building data-intensive applications, interactive visualization tools, and operational software for geospatial intelligence, SaaS, healthcare, and retail organizations. Experienced technical lead specializing in React, TypeScript, FastAPI, and real-time application architecture, with a passion for transforming complex systems and workflows into intuitive user experiences.

## SKILLS

- Frontend: TypeScript, React.js, Next.js, Tailwind, Redux, RTK Query, PWA, WebGL
- Geospatial: Mapbox GL, OpenLayers, Cesium/Resium, Raster/Vector Tiles, PostGIS, GeoJSON
- Backend: Node.js, FastAPI (Python), PostgreSQL, Supabase, WebSockets
- Infrastructure: GitHub Actions, Vercel, Render

## PROFESSIONAL EXPERIENCE

### Senior Software Engineer | EarthDaily Federal – Geospatial Defense Contractor | 03/2025 – 02/2026 (1 Year Contract)

- Led frontend development of a geospatial search-and-rescue platform for the Air Force Rescue Coordination Center, replacing a paper-based workflow for distress beacon triage and mission management, improving speed and clarity in time-critical rescue operations. The system was iteratively refined through direct operator feedback during live beta deployment, strengthening coordination across multiple agencies.
- Architected an AI-assisted command interface using React and FastAPI that translated natural language prompts into structured geospatial UI actions through a dynamic tool registry.
- Built a Mapbox-based geospatial intelligence interface in collaboration with Army analysts, designed to integrate machine learning models for change detection and object detection (e.g., airstrips and vessel movement) and OSINT data tied to areas of interest.
- Developed an interactive geospatial annotation workspace supporting map labeling, callouts, and composable overlays, including drag-and-drop widgets such as overview maps and picture-in-picture views for comparative analysis.
- Proposed and prototyped a report generation workflow that was adopted into the product roadmap, enabling analysts to compile images and intelligence artifacts from map and annotation tools into structured reports with live PDF preview and in-place editing.
- Tech Stack: React, TypeScript, OpenLayers, Mapbox GL, FastAPI (Python)

### Lead Frontend Software Engineer | Stellar/Storable – Marina Management Software | 02/2021 – 05/2024

- Led frontend development across two products serving marina operators and customers, coordinating a 4-person engineering team (including offshore developers) and partnering directly with the CTO on product delivery and technical direction.
- Developed multi-tenant web applications using React and TypeScript, supporting booking, sales, and customer management workflows across separate PostgreSQL-backed systems.
- Built a Progressive Web Application (PWA) enabling dock attendants to continue operations in low-connectivity environments while synchronizing data when connections were restored.
- Designed and implemented an interactive marina map with live berth availability, drag-and-drop slip assignments, and automated contract updates, reducing manual administrative work for marina staff.

### Software Engineer | Enterprise & Early Career Experience | 2015 – 2020

- Built React, Angular, and legacy JavaScript applications across healthcare, retail e-commerce, and restaurant POS systems, supporting enrollment workflows, product and cart experiences, and distributed device-based inventory operations.
- Developed a WebSocket-based cloud POS and inventory system enabling real-time communication across distributed restaurant devices using AngularJS.
- Collaborated with UX and engineering teams to improve usability, performance, and delivery velocity across multiple production systems, including insurance enrollment and retail platforms such as BestBuy.com.

## EDUCATION

B.S. Mathematical Physics – Saint Cloud State University

A.S. Biomedical Technology – Anoka-Ramsey Community College